

CHApel MultiPhysics Simulation Software (CHAMPS)

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**POLYTECHNIQUE
MONTREAL**

TECHNOLOGICAL
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CHAMPS – Modeling Configuration

Model / Method	NACA0012	ONERA M6
Turbulence Model	Spalart–Allmaras (SA)	Spalart–Allmaras (SA)
Heat-Transfer Method	Recovery Temperature Approximation	Recovery Temperature Approximation
Droplet Model	Eulerian	Eulerian
Droplet Drag Model	Deformable Sphere	Deformable Sphere
SLD Corrections	None	None
Thermodynamic Model	Messinger	Messinger
Ice Density	Constant	Constant
Surface Roughness	Constant	Constant
Geometry Evolution	Algebraic (Single-shot) / Level Set (Multilayer)	Algebraic (Single-shot) / Level Set (Multilayer)
Accretion Strategy	Single-shot & 5 Layers	Single-shot & 5 Layers



CHAMPS – Multi-Fidelity and Multiphysics Capabilities

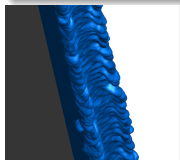
Multi-Fidelity Aerodynamics

NL-VLM → Full Potential → Euler → RANS → LES
→ DNS

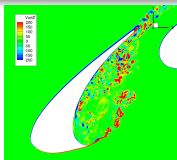
Multiphysics Capabilities

Fluid-Structure Interaction (FSI)
Contrails

• Icing •



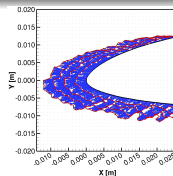
Deterministic Icing



Scale-Resolving
CFD (DDES)



Contrails*



Stochastic Icing

FSI
Figure Placeholder

Fluid-Structure
Interaction

* Contrail modeling developed in collaboration with Prof. Roberto Paoli.



CHAMPS: Advanced 2D-3D CFD Solver

CHAMPS (CHapel MultiPhysics Software)

- 2D-3D Unstructured Reynolds Average Navier Stokes Solver
- Second order finite volume
- Convective Fluxes : Roe, AUSM
- SA, $k-\omega$ SST-V and Langtry-Menter transitional turbulence models
- Explicit solver (Runge-Kutta) and implicit solvers (SGS, GMRES)
- Linked to external libraries: MKL, CGNS, METIS and PETSC
- Multi-Fidelity Solvers: Potential, Non Linear Vortex Lattice, Euler, RANS, URANS, LES, DNS
- Multi-Physics: Icing (Deterministic, Stochastic), Fluid-Structure Interactions, Contrails

CHAMPS Software : 150,000 lines of code !

- Pre-Processor $\approx$ 17K lines
- Post Processor $\approx$ 2K lines
- Flow Solver $\approx$ 15K lines
- Turbulence Solver $\approx$ 13K lines
- Droplet Solver (Eulerian + Lagrangian) $\approx$ 24K lines
- Fluid-Structure Interaction Solver (Coupling + FEM) $\approx$ 15K lines